

SUNGHEE YUN
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SUMMARY

I am Co-Founder & CTO @ Erudio Bio, Inc., CA, USA, Co-Founder & Leader of Silicon Valley AI Nexus (K-PAI), a KFAS-Salzburg Global Leadership Initiative Fellow @ Salzburg Global Seminar, Salzburg, Austria, AI-Korean Medicine Integration Initiative Task Force Member @ The Association of Korean Medicine, Seoul, South Korea, a Visiting Professor @ Sogang University, Seoul, South Korea, an Advisory Professor @ Daegu Gyeongbuk Institute of Science & Technology (DGIST), South Korea, Global Advisory Board Member @ Innovative Future Brain-Inspired Intelligence System Semiconductor of Sogang University, and Network Expert @ Gerson Lehrman Group, Inc..

Before founding the new AI biotech company, I co-founded Gauss Labs, Inc., an industrial AI company, built and ran the US Headquarters of Gauss Labs, spearheaded research & development of the core technology & products. Before Gauss Labs, I drove various AI technology and product developments and productionizations in the field of e-Commerce @ Amazon.com, Inc. where one of the projects created a 200 USD revenue increase via the Amazon Mobile Shopping App. Before Amazon, I worked for Software Research Center, Strategic Sales & Marketing Team, Design Technology Team & Computer-Aided Engineering (CAE) Team of Samsung Semiconductor, Inc. where I developed diverse AI and optimization tools for semiconductor chip designers, manufacturing engineers & test engineers where more than 200 engineers still use the circuit optimization tools and generic AI optimization platform, dubbed iOpt, every day.

I received a BS degree in Electrical Engineering (EE) from Seoul National University (SNU), Seoul, South Korea and MS & Ph.D. degrees in EE from Stanford University, CA, USA under supervision of Prof. Stephen Boyd, who is a legendary figure in the field. (Having him as my advisor was one of the luckiest things that have ever happened to my life!)

I specialize in Artificial Intelligence (AI) and Convex Optimization leading cutting-edge projects across diverse sectors including semiconductor design, strategic marketing with market intelligence, e-commerce, industrial AI, and biotech while driving business development, connecting people, and consulting people in diverse industries. My focus is on leveraging the AI technology especially around AI agents based on recent breakthroughs of LLMs and multimodal genAI to drive business innovation and predict industry trends.

As the Leader of Silicon Valley AI Nexus (K-PAI), I foster collaboration between industry leaders, researchers, and policymakers to advance responsible AI technologies and establish ethical frameworks for AI deployment across critical sectors including healthcare, finance, and biotechnology. K-PAI serves as a vital bridge connecting cutting-edge AI research with real-world applications, where my experience at companies like Erudio Bio, Gauss Labs, Amazon, and Samsung provides unique insights into the practical challenges of implementing secure and trustworthy AI solutions at scale. Through K-PAI's seminars, workshops, and policy discussions, I work to ensure that as AI systems become more pervasive, they are built with responsible design principles that balance innovation with societal benefits, particularly addressing the complex regulatory landscapes across the US, Europe, and Asia-Pacific regions where different technological standards and cultural expectations around AI governance create both challenges and opportunities for sustainable AI development that serves humanity's long-term interests.

As a KFAS-Salzburg Global Leadership Initiative Fellow, I develop actionable solutions for the challenges posed by a rapidly changing world by AI. My work focuses on "Technology, Growth, and Inequality: The Case of AI," within the broader theme of "Uncertain Futures and Connections Reimagined: Connecting Technologies." I also actively contribute to AI and semiconductor technology policy making, advising state and national governments while considering the geopolitical dynamics of key players like the US, China, Japan, Taiwan, and South Korea.

I also dedicate to fostering knowledge, insight, and experience transfer and innovation by serving as a Visiting Professor of Department of Electronic Engineering @ Sogang University, Seoul, South Korea, an Advisory Professor of Department of Electrical Engineering & Computer Science (EECS) @ Daegu Gyeongbuk In-

stitute of Science & Technology (DGIST), South Korea, and Global Advisory Board Member @ Innovative Future Brain-Inspired Intelligence System Semiconductor of Sogang University.

My life goal is to help build a future where humanity thrives through the advancement of health, safety, liberty, and equality—foundational elements that pave the way to genuine happiness.

EXPERIENCE

Co-Founder & CTO Erudio Bio, Inc.	Dec-2023 – Present
AI Technology, and Product Strategy to develop and productionize AI-enhanced multiplexing bio data platform	
Co-Founder & Leader Silicon Valley AI Nexus (K-PAI)	Oct-2024 – Present
Advisor Korean American Semiconductor Professional Alliance (KASPA)	Jan-2026 – Present
CGO / Global Managing Partner LULUMEDIC	Nov-2025 – Present
KFAS-Salzburg Global Leadership Initiative Fellow Salzburg Global Seminar	Aug-2024 – Present
Address the challenges posed by our rapidly changing world in the working group, “Technology, Growth, and Inequality: The Case of AI” under the bigger theme “for Uncertain Futures and Connections Reimagined: Connecting Technologies!”	
AI-Korean Medicine Integration Advisor The Association of Korean Medicine	Aug-2025 – Present
Visiting Professor Electronic Engineering Department, Sogang University	Mar-2020 – Present
Advisory Professor Information and Communication Engineering, DGIST	Feb-2020 – Present
Global Advisory Board Member Innovative Future Brain-Inspired Intelligence System Semiconductor of Sogang University	Aug-2020 – Present
Consult on academic direction, industry trend in Silicon Valley, social scientific / ethical / legal issues of AI & special lectures and seminars	
Network Expert Gerson Lehrman Group (GLG), New York City, U.S.A.	Feb-2022 – Present
Help clients solve their toughest business and technical challenges around manufacturing profit and sales improvement, e-commerce customer engagement level improvement, and AI technology in biotech industry	
Co-Founder & CTO - Senior Fellow Gauss Labs, Inc., Palo Alto, CA. U.S.A	May-2000 – Nov-2023
VP - Fellow SK hynix	Sep-2020 – Jun-2021
Drive technology innovation and business strategies & lead Gauss Labs US Headquarters in Silicon Valley	
Senior Applied Scientist Amazon.com, Inc., Vancouver, BC, Canada	Nov-2018 – Aug-2020
- AI for Mobile Shopping App Main Menu Personalization system creating revenue by 200MM USD - TestBot AI agents patching and updating software test scenarios	
Principal Engineer Software R&D Center, Samsung Electronics	Dec-2016 – Jun-2017
Decentralized Federated AI for Internet of Things (IoT)	
Principal Engineer Strategic Sales & Marketing Team, Samsung Electronics	Dec-2015 – Dec-2016
AI for Sales Forecasting of world-wide NAND Flash and DRAM markets	
Principal Engineer Design Technology Team, Samsung Electronics	Mar-2012 – Nov-2015
- AI for Optimal Scheduling System for Test & Package Factories in Suzhou, China, - Generic AI & Optimization Platform for automatization of semiconductor design & manufacturing	

Senior Engineer CAE Team, Semiconductor R&D Center, Samsung Electronics Dec-2004 – Feb-2012

- NAND Flash Memory Core Operation Optimization achieving 10% accuracy improvement
- Generic Digital/Analog/Mixed-signal Circuit Optimization
- High density DRAM failure rate estimation achieving > 100X prediction time reduction

Research Assistant EE Department, Stanford University, CA, U.S.A. Mar-1999 – Nov-2004

- Multi-objective Logic Circuits Design using Convex Optimization

Development Engineer Voyan Technology, Santa Clara, CA, U.S.A. Sep-2000 – Aug-2001

- Digital Subscriber Line (DSL) Diagnosis & Length Adjustable Transmission Line Emulator

EDUCATION

Ph.D., Electrical Engineering, Stanford University, CA, U.S.A. Sep-2001 – Nov-2004

M.S., Electrical Engineering, Stanford University, CA, U.S.A. Sep-1998 – Jun-2000

B.S., Electrical Engineering, Seoul National University, Seoul, Korea Mar-1994 – Feb-1998

HONORS AND AWARDS

- Best Invention Award at Amazon.com, Inc. 2019
- Best Paper Award at Samsung Semiconductor Technical Paper Award 2011
- Best Paper Award at Samsung Group Best Paper Award 2005
- Best Paper Award at International Symposium on Quality Electronic Design (ISQED) 2005
- Full Scholarship from Samsung Electronics 2001–2004
- Samsung Frontier Membership 1999
- Ranked 2nd in the Ph.D. Qualifying Examination, Stanford University 1999
- Full Scholarship from Korean Foundation for Advanced Study (KFAS) 1998–2000
- Graduated with Honors - Magna Cum Laude, Seoul National University (SNU) 1998
- Full Scholarship from Seoul National University (SNU) 1994–1998
- Ranked 2nd in University Entrance Examination, Seoul National University (SNU) 1994
- Graduated with Honors - Summa Cum Laude, Seoul Science High School 1993
- Bronze Medal at Asian Pacific Mathematical Olympiad (APMO) 1992
- Silver Medal at Korea Mathematical Olympiad (KMO) 1992

INVITED SEMINARS, LECTURES, AND CONSULTATIONS

- “[Tech University of Korea Special AI Lecture] The Complete AI Picture - Technology Breakthroughs, Industry Transformation, Market Forces, and Strategic Investment Insights” invited by Tech University of Korea 02-Apr-2026 KST
- “[SNUSV.NET Invited Lecture for Entrepreneurs] From Amazon AI to Gates Foundation Biotech - Three Pivots, Two Countries, One Mission in the Bio-Medical Revolution” invited by Seoul National University Student Venture Network (SNUSV.NET) 30-Mar-2026 KST
- “[SNU College of Engineering Special AI Lecture] The AI Revolution for Engineers - From Algorithms to Agentic Systems, Silicon Valley to Global Impact” invited by Seoul National University (SNU) College of Engineering 28-Mar-2026 KST
- “[MIT GSW 2026 Special AI Lecture] The \$1 Trillion AI-Biotech Revolution - Technology Foundations, Market Dynamics, and Why This Convergence Will Create the Next Unicorns” invited by MIT Global Startup Workshop (GSW) 27-Mar-2026 KST

- “[Hoban Science Bridge] Town Hall Meeting Panel Discussion” invited by K-Science Academy 26-Mar-2026 KST
- “[Hwarang SV District Vision Festival VIP Guest Speech] Living with Integrity over Vision in the AI Era - Building Competitive Excellence Through Domain Expertise with Moral Compass” invited by Hwarang Youth Foundation Silicon Valley District 22-Feb-2026 PST
- “[Special AI Lecture for Legal Professionals] The AI Paradigm Shift for Legal Minds - Silicon Valley Insights, Cognitive Science Foundations, and the Intersection of Logic and Pattern Recognition” invited by K-PAI Legal Professionals Group @ Stanford University 21-Feb-2026 PST
- “[KSEA Special AI Lecture] Beyond ChatGPT - The Complete AI Landscape of Technology, Hardware, Markets, and Bridging Research to Startup Reality” invited by Korean-American Scientists and Engineers Association (KSEA), Austin Chapter, Austin, Texas 13-Feb-2026 CST
- “[Texas A&M Special AI Lecture] AI Technology, Trends, and Market - Industrial AI” invited by Texas A&M University 13-Feb-2026 CST
- “[San Jose State University Special AI Lecture Series VIII - RL/Recent Progress/App Development] From Reinforcement Learning to Production - RL, Recent Breakthroughs & App Development” invited by San José State University 16-Jan-2026 PST
- “[San Jose State University Special AI Lecture Series VII - ML Fundamentals/Vibe Coding] ML Theory Meets Practice - Frameworks, Backpropagation & Live Coding” invited by San José State University 15-Jan-2026 PST
- “[K-PAI’s 14th Chapter: 2026 New Year Celebration - Reflecting, Connecting, and Celebrating] The holistic synthesis of AI - The AI Landscape through K-PAI’s Lens” invited by K-PAI 14-Jan-2026 PST
- “[San Jose State University Special AI Lecture Series VI - Prerequisites/Learning Strategies/LLM Tools] How to Master AI/ML - Prerequisites, Study Strategies & LLM as Your Learning Partner” invited by San José State University 14-Jan-2026 PST
- “[ABDD Summit 2026] iTalk - Erudio Bio & Silicon Valley AI Forum (K-PAI)” invited by K-BioX 11-Jan-2026 PST
- “[San Jose State University Special AI Lecture Series V - Industrial AI from First-Hand Experience] From Theory to Production - Manufacturing AI Systems, Computer Vision, Virtual Metrology, and Project Success Strategies” invited by San José State University 01-Jan-2026 PST
- “[San Jose State University Special AI Lecture Series IV - Philosophy, Ethics & Consciousness] AI’s Philosophical Frontiers - Consciousness, Knowledge, Belief, Ethics, and Limits of Machine Reasoning” invited by San José State University 01-Jan-2026 PST
- “[San Jose State University Special AI Lecture Series III - AI + Biotech & Physical AI] The Convergence Revolution - AlphaFold 3, AI-Driven Biotech, Humanoid Robots, and Future of Physical Intelligence” invited by San José State University 31-Dec-2025 PST
- “[San Jose State University Special AI Lecture Series II - LLM & GenAI Deep Dive] Inside the Generative Revolution - Transformers and the Technology-Market Nexus of LLMs” invited by San José State University 30-Dec-2025 PST
- “[San Jose State University Special AI Lecture Series I - The Historical Foundation & Modern Landscape] From Connectionism vs Symbolism to the Agentic Revolution - Understanding Deep Learning, LLMs, and Path to Multi-Modal Intelligence” invited by San José State University 29-Dec-2025 PST
- “[AI-BioX ConfEX Grand Summit 2025] The Real Silicon Valley AI Story - A Critical K-PAI Insider’s Guide to Where Technology Is Truly Heading” invited by AI-BioX ConfEX Summit 12-Dec-2025 KST
- “[SNU ECE AI Fireside Chat] Finding Your Path Through the Complete Landscape - A Practitioner’s Journey from Theory to Impact” invited by Department of Electrical and Computer Engineering (ECE), Seoul National University (SNU) 26-Nov-2025 KST

- “[AI & Entrepreneurship Consultation for KMEA] Navigating the AI-Bio Frontier - Technical Strategy and Entrepreneurial Decision-Making for MDs” invited by Korean Medical Entrepreneurship Association (KMEA) 22-Nov-2025 KST
- “[Jeju Bio Forum Special AI Seminar] The AI Recipe for Food Innovation - Silicon Valley Technologies, Business Models, and Opportunities for FoodTech” invited by Jeju Biotechnology Business Association & Jeju Technopark 21-Nov-2025 KST
- “[Korea University Special AI Lecture] AI-Powered Drug Discovery and Diagnostics - Technology, Business Models, and the Entrepreneurial Journey” invited by Korea University College of Medicine 19-Nov-2025 PST
- “[Future of Medicine] From Pixels to Patients - How Agentic AI Transforms Pathology’s Data Bottleneck” invited by K-BioX 18-Nov-2025 KST
- “[Strategic Consultation on Bio-Medical Conferences] Architecting World-Class AI-Bio Convergence Summits - Strategic Advisory for G-Quantum Bio Annual Conferences” sought by Raon Data & Consulting 18-Nov-2025 KST
- “[K-PAI - IITP Roundtable Networking Dinner] The AI Ecosystem - A Silicon Valley Field Guide to Technology, Markets, and Meaning” invited by Institute of Information & Communications Technology Planning & Evaluation (IITP) 15-Nov-2025 PST
- “[Special AI Lecture for Hwarang Friends] Becoming the Leader of AI Era - From ChatGPT to Your Future Superpower” invited by Hwarang International 15-Nov-2025 PST
- “[Special AI Lecture for Hwarang friends] AI Awakens: From ChatGPT to Your Future Superpowers” invited by Hwarang International 15-Nov-2025 PST
- “[IFEZ X K-BioX - ABDD Summit] The Holistic AI-Biotech Landscape - Silicon Valley Technologies Reshaping Drug Discovery and Precision Medicine” invited by Incheon Free Economic Zone (IFEZ) and K-BioX 07-Nov-2025 KST
- “[KUBS - Center for Digital Transformation & Business (CDTB)] The Economics of Intelligence - How AI is Reshaping Business Models, Markets, and Competitive Advantage” invited by Center for Digital Transformation & Business (CDTB) of Korea University Business School (KUBS) 24-Oct-2025 KST
- “[DGIST Special AI Seminar] From LLMs to Agentic AI - Understanding the Paradigm Shift and Its Implications for Technology and Society” invited by Department of Electrical Engineering and Computer Science (EECS) of Daegu Gyeongbuk Institute of Science & Technology (DGIST) 22-Oct-2025 KST
- “[Hanyang Univ Special AI Lecture] Innovation Leadership in the AI Era - Knowledge, Experience, and Insight for Tomorrow’s Technology Managers” invited by Hanyang University Graduate School of Technology & Innovation Management 21-Oct-2025 KST
- “[SEOULTECH Special AI Lecture] Beyond ChatGPT - Navigating AI Agents, Biotech Convergence, and Global Transformation” invited by Seoul National University of Science and Technology (SEOULTECH) 20-Oct-2025 KST
- “[ICTC 2025 AI Seminar] Convergent AI Innovation - From Protein Folding Breakthroughs to Agentic AI for Sustainable Future” invited by International Conference on ICT Convergence (ICTC) 2025 15-Oct-2025 KST
- “[AI Special Lecture for Jihoo’s Family] Convergent AI Innovation - From Protein Folding Breakthroughs to Agentic AI for Sustainable Future” invited by Jihoo’s Family 12-Oct-2025 KST
- “[Silicon Valley Korean Professionals Roundtable AI Special Lecture on AI, Semiconductors, and Biotechnology] The True AI Revolution - Essential Insights for Policymakers on Technology, Economy, and Human Impact” invited by KOTRA Silicon Valley 07-Oct-2025 PDT
- “[KICSV Special AI Lecture] Mathematics for AI - Theory into Practice” invited by Korea Innovation Center Silicon Valley (KICSV) 06-Oct-2025 PDT

- “[KICSV Special AI Lecture] Core AI Concepts and Modern Architectures” invited by Korea Innovation Center Silicon Valley (KICSV) 06-Oct-2025 PDT
- “[Korean Medicine AI Integration Task Force Meeting] From Silicon Valley to Biotech Valley - How Agentic AI is Reshaping Drug Discovery” invited by The Association of Korean Medicine 26-Sep-2025 KST
- “[G-BIVA Innovation Salon] Intelligent Agents, Intelligent Medicine - AI’s Role in Next-Gen Healthcare” invited by Suwon City KSEA SV K-BioX Summit 8 23-Sep-2025 KST
- “[G-Quantum Bio Summit 2025] From Silicon Valley to Biotech Valley - AI Agents Reshaping the Future of Drug Discovery and Precision Medicine” invited by Suwon City KSEA SV K-BioX Summit 8 23-Sep-2025 KST
- “[SNU ECE Four-Departments Joint Seminar] AI Pulse Check - What Silicon Valley Is Actually Building Right Now” invited by Department of Electrical and Computer Engineering (ECE), Seoul National University (SNU) 18-Sep-2025 KST
- “[KUBS Special AI Lecture] The AI Revolution - How Smart Systems Are Reshaping Industries and Executive Decision-Making” invited by Korea University Business School 18-Sep-2025 KST
- “[POSTECH Creativity IT Seminar] The AI Landscape Decoded - Where We Are, Where We’re Going” invited by Department of Convergence IT Engineering, POSTECH 17-Sep-2025 KST
- “[2025 Middle School Gifted Education Dream Realization Camp] The Magic Behind the Machine - AI as Your Creative Partner” 12-Aug-2025 KST
- “[BSR Korea Special AI Lecture] Silicon Valley’s AI Revolution Meets Biotech - Technology, Markets, and the Future of Drug Discovery” invited by BSR Korea 08-Aug-2025 KST
- “[KAIST Medical Science and Engineering Special AI Lecture] Leading AI Innovation Across Industries - Silicon Valley Insights on Technology Transfer from Semiconductor & E-Commerce to AI-powered Biotech” invited by Graduate School of Medical Science and Engineering, Korea Advanced Institute of Science & Technology (KAIST) 03-Jul-2025 KST
- “[Institut Pasteur Korea Special AI Lecture] Silicon Valley AI Innovation - Silicon Valley Insights on Technology Transfer from Semiconductor & E-Commerce to AI-Powered Biotech” invited by Institut Pasteur Korea 02-Jul-2025 KST
- “[Yonsei University Applied Statistics AI-Biotech Seminar] The Era of Agentic AI - From Deep Learning and LLMs to Biotech Breakthroughs” invited by Department of Applied Statistics and Department of Statistics and Data Science, Yonsei University 01-Jul-2025 KST
- “[Ewha Womans University AI-Biotech Seminar] Building Tomorrow’s Healthcare - LLM, genAI, Agentic AI, and Human-Centered Innovation” invited by Division of Mechanical and Biomedical Engineering, Ewha Womans University 01-Jul-2025 KST
- “[KIST AI Policy Seminar] Cross-Industry AI Innovation - Silicon Valley’s Model for AI-Driven Technology Transfer Across Industries” invited by Korea Institute of Science & Technology (KIST) 30-Jun-2025 KST
- “[KFAS 2025 - AI & Humanity] The AI Metamorphosis - Navigating New Frontiers in Technology, Society, and Human Experience” invited by Knowledge Festival All Seasons (KFAS) 2025 27-Jun-2025 KST
- “[2025 KAIST BBE Bay Area Research Exchange] Panel Discussion on Global Biotech Innovation Ecosystem” @ Stanford University invited by Korea Advanced Institute of Science & Technology (KAIST) Bio & Brain Engineering (BBE), Korea Innovation Center (KIC), Korean Life Scientists in the Bay Area (KOLIS), and Stanford K-BioX 22-Jun-2025 PDT
- “[KASBP AI-biotech Seminar] Beyond AlphaFold - Building AI-Powered Biomarker Platforms for the Next Generation of Drug Discovery” invited by Korean American Society in Biotech and Pharmaceuticals (KASBP-SF) 04-Jun-2025 PDT
- “[KIC Silicon Valley Innovation Workshop] Silicon Valley’s Cultural Engine of Innovation and Disruption” invited by Korea Innovation Center (KIC) Silicon Valley Membership Innovation Workshop & Collaboration Networking 2025 23-May-2025 KST

- “[Korea University Business School AI Seminar] The AI Paradigm Shift - Technological Foundations, Market Dynamics, and Human Impact” invited by Korea University Business School 22-May-2025 KST
- “[Saramin Executive AI Consultation] AI Technology and Business Strategy Consultation” invited by saramin 08-May-2025 KST
- “[Sungkyunkwan University AI Seminar] AI Renaissance - New Paradigms of Knowledge, Business Innovation, and Social Architecture” invited by College of Information and Communication Engineering of Sungkyunkwan University 07-May-2025 KST
- “[SNU INMC AI Seminar] The AI Landscape: Navigating Technology, Industry Shifts, and Future Trends” invited by SNU Institute of New Media and Communications (INMC) 07-May-2025 KST
- “[Korea University AI Seminar] The AI Architecture Decoded - Orchestrating Technology and Hardware Innovation” invited by Korea University 02-May-2025 KST
- “[POSTECH AI Seminar] The AI Renaissance 2.0 - Redefining Knowledge Systems and Human-Machine Collaboration” invited by Department of Electrical Engineering of POSTECH 01-May-2025 KST
- “[Seoul Science High School AI Seminar] AI - Technology, Industry, Market, Humanities, and Your Future!” invited by Seoul Science High School 29-Apr-2025 KST
- “[Samsung Semiconductor Insight-Sharing Session] Insights from Silicon Valley - Innovation, Leadership, and AI in the Global Tech Ecosystem” invited by Foundry CMP Technology Team, Samsung Semiconductor, South Korea 05-Mar-2025 KST
- “[Samsung Semiconductor AI Seminar] AI - Technologies, Industry Transformations, Societal Impacts, and Future Trajectories with Focus on Semiconductor Manufacturing Applications” invited by Foundry CMP Technology Team, Samsung Semiconductor 05-Mar-2025 KST
- “[Samsung AI Seminar] AI - Technologies, Industry Transformations, Societal Impacts, and Future Trajectories with Focus on Semiconductor Manufacturing Applications” invited by Foundry CMP Technology Team, Samsung Semiconductor 05-Mar-2025
- “[KAIST AI Seminar] AI in Action: Biotechnology and Industrial Applications” invited by Department of Mechanical Engineering @ Korea Advanced Institute of Science and Technology (KAIST) 28-Feb-2025
- “[Sogang University BK Intelligence System Semiconductor Invited AI Seminar] The AI Metamorphosis: From Silicon Valley to Global Society - Technology, Consciousness, and Human Impact” invited by Sogang University 26-Feb-2025
- “[K-PAI Seminar] The AI Knight Rises - From Deep Learning to Flourishing Societies” invited by Silicon Valley Korea Private AI Forum 22-Jan-2025
- “[Christmas Seminar] The AI Knight Rises - From Deep Learning to Flourishing Societies” hosted by Jisun in New York 27-Dec-2024
- “[CryptoLab Lyon Research Center Seminar] AI - Technology, Industry & Hardware” invited by CryptoLab Lyon Research Center, Lyon, France 02-Dec-2024 CET
- “[KAIST AI Seminar] AI - Technology, Industry, Market & Hardware” invited by Department of Mechanical Engineering, Korea Advanced Institute of Science and Technology (KAIST) 04-Nov-2024 KST
- “[Solidigm Executive AI Consultation] Technical & Business AI Strategy Consultation for Data Center Flash Storages” invited by Solidigm CEO 30-Oct-2024 PDT
- “[Texas A&M Seminar / Intro to Semiconductor & Microelectronics] Industrial AI - Technological Innovations and Impacts on Semiconductor Manufacturing” invited by Texas A&M University 18-Oct-2024
- “AI for Fusion Biweekly Seminar: Industrial AI & Biotechnology - Technology, Market, and Future” invited by EnableFusion 08-Aug-2024
- “Applied Statistics Invited Seminar LLM & genAI - Technology, Industry, and Market” invited by Yonsei University 10-Jul-2024

- “Samsung Flash Design Team Invited Seminar - Industrial AI & its applications in manufacturing” invited by Samsung Semiconductor 27-Jun-2024
- “SNU CSE Invited Seminar - LLM & genAI - Technology, Industry, and Some Important Questions” invited by Seoul National University (SNU) 25-Jun-2024
- “BK Intelligence System Semiconductor Invited Seminar - LLM & genAI - Technology, Industry, and Some Important Questions” invited by Sogang University 24-Jun-2024
- “2024 LINC 3.0 + GSAI Joint Seminar - Industrial AI - Best Practices in Semiconductor Manufacturing” invited by Pohang University of Science and Technology (POSTECH) 19-Jun-2024
- “SAIT AIRC Seminar: LLM & getnAI - Technology, Business, and AI Markets” invited by Samsung Advanced Institute of Technology (SAIT) 24-Apr-2024
- “DGIST Seminar: LLM & getnAI - Technology, Business, and AI Markets” invited by EE Department of Daegu Gyeongbuk Institute of Science and Technology (DGIST) 06-Feb-2024
- “Cryptography Seminar: Industrial AI Technology in Manufacturing” invited by SNU Mathematics Department 02-Feb-2024
- “SNU Data Science (DS) Seminar: Industrial AI Technology” invited by Seoul National University (SNU) Data Science (DS) Graduate School 01-Nov-2023
- “Time-series Learning and Software Platform for Manufacturing AI” invited by EE Department of Daegu Gyeongbuk Institute of Science & Technology (DGIST) 23-Jun-2022
- “BK4 Seminar: Manufacturing AI Problems and Solutions” invited by Sogang University 13-Jun-2022
- “EE/CS Seminar: Manufacturing AI Problems and Solutions” invited by SNU EE/CS Departments 13-Jun-2022
- “Time-series Machine Learning in Manufacturing” invited by POSTECH 08-Apr-2022
- “AI Frontier Summit: AI in Manufacturing and Industry Areas” invited by Korean Institute of Communications and Information Sciences (KICS) Conference 2021
- “Special Lecture: Electrical Engineer in the age of AI” invited by Sogang University 26-Mar-2021
- “Future Creation Seminar: The Future of AI” invited by Amorepacific 20-Oct-2020
- “Introduction to AI and Machine Learning” invited by Consulate General of Republic of Korea @ Vancouver 17-May-2019
- “Reinforcement Learning and its Applications” invited by University of British Columbia (UBC) Sauder School of Business 26-Apr-2019
- “Convex Optimization for Machine Learning (ML)” invited by SNU EE Department 22-Aug-2018
- “Convex Optimization for Artificial Intelligence (AI)” invited by Amazon Machine Learning (ML) Meetings 02-Feb-2018
- “KCC Tutorial: Convex Optimization for Decentralized Machine Learning” invited by Korea Computer Congress (KCC) 2017 18-Jun-2017
- “Convex Optimization for Machine Learning” invited by Samsung Semiconductor DRAM Development Lab 2017
- “Convex Optimization for Machine Learning” invited by EE Department of Daegu Gyeongbuk Institute of Science & Technology (DGIST) 2017
- “Convex Optimization for Machine Learning” invited by EE Department of Ulsan National Institute of Science & Technology (UNIST) 2017
- “Introduction to Convex Optimization and its Applications” invited by Kyung Hee University 2013
- “Convex Optimization and its Applications” invited by Seoul National University (SNU) 2008
- “Circuit Design via Geometric Programming” invited by Seoul National University (SNU) 2005
- “Convex Optimization and its Application in Circuit Design” invited by Samsung Semiconductor Memroy Business Computer-aided Engineering (CAE) Team 2002

PUBLICATIONS

- Sangyoun Lee, Juho Jung, Changdae Oh, and Sunghye Yun. “Enhancing Temporal Action Localization: Advanced S6 Modeling with Recurrent Mechanism”, <https://arxiv.org/abs/2407.13078>, 2024
- S. Zabrocki, P. Jo, C. Park, D. Yim, S. Yun, and B. Lee. “Adaptive Online Time-Series Prediction for Virtual Metrology in Semiconductor Manufacturing”, 34th Annual SEMI Advanced Semiconductor Manufacturing Conference (ASMC), May 2023
- M. Karpusha, S. Yun, and I. Fehervari. “Calibrated neighborhood aware confidence measure for deep metric learning”, (<https://arxiv.org/abs/2006.04935>), June 2020
- J. Lee, S. Yun, J. Kim, D. Kang, J. Kim, and S. Lee. “Multiple shift-vector importance sampling method using support vector machine and clustering for high-density DRAM designs”, 17th International Symposium on Quality Electronic Design (ISQED), pp. 430–436, March 2016
- J. Jeon, I.H. Park, M. Kang, W. Hahn, K. Choi, S. Yun, G. Yang, K. Lee, Y. Park, and C. Chung. “Accurate compact modeling for sub-20-nm NAND Flash cell array simulation using the PSP model”, IEEE Transactions on Electron Devices, Vol. 59, No. 12, pp. 3503–3509, December 2012
- S. Yun, I.H. Park, N.H. Lee, and J.K. Jeong. “New design methodology for efficient NAND Flash development using multiobjective evolutionary algorithm”, Samsung Semiconductor Technical Paper Award, August 2011 (Best Paper Award)
- S.-J. Kim, S. Boyd, S. Yun, D. Patil, and M. Horowitz. “A heuristic for optimizing stochastic activity networks with applications to statistical digital circuit sizing”, Optimization and Engineering, Vol. 8, No. 4, pp. 397–430, December 2007
- S. Yun, D.H. Kim, and S.H. Jung. “Statistical parameter estimation methodology for statistical circuit analysis”, Samsung Group Paper Award, November 2005 (Best Paper Award)
- D. Patil, S. Yun, S.-J. Kim, A. Cheung, M. Horowitz, and S. Boyd. “A new method for design of robust digital circuits”, 6th International Symposium on Quality Electronic Design (ISQED), San Jose, pp. 676–681, March 2005 (Best Paper Award)
- S. Boyd, L. Vandenberghe, A. El Gamal, and S. Yun. “Design of robust global power and ground networks”, Proceedings ACM/SIGDA Symposium on Physical Design (ISPD), pp. 60–65, April 2001

PATENTS

- | | |
|---|------|
| • Systems and Methods for Process Monitoring and Control | 2024 |
| • Testbot AI agent: automatic software test scenario patch and modification system | 2020 |
| • Method of testing semiconductor memory device, test device, and computer readable recording medium for recording test program for semiconductor memory device | 2017 |
| • Method and system for designing 3D semiconductor package | 2014 |
| • Methodology for estimating statistical distribution characteristics of physical parameters of semiconductor device | 2009 |
| • Methodology for estimating statistical distribution characteristics of product parameters | 2007 |
| • Length adjustable transmission line emulators with simple RLC circuits | 2001 |

TEACHING EXPERIENCE

- | | |
|--|-----------|
| Amazon Lecturer Machine Learning University, Amazon | 2018–2020 |
| • AI Mathematical Fundamentals, AI Data Science, Advanced Machine Learning Courses | |
| Samsung Electronics Lecturer Samsung Electronics | 2005–2017 |
| • Convex Optimization for AI, Machine Learning & AI | |
| • Numerical Linear Algebra for ML, Advanced Statistics | |
| Teaching Assistant Stanford University | 1999–2002 |
| • Linear Dynamical Systems, Convex Optimization | |